Lecture 5

Casimir operators and complete reducibility

Objectives. *Lecture 3 gave a module with an invariant line and no invariant complement. We prove that this cannot happen for a semisimple Lie algebra. The tool is the Casimir operator, built from the trace form of the module.*

By the end of this lecture you will be able to:

- Prove that for semisimple $\mathfrak{g}$ the trace form is nondegenerate on the faithful quotient.
- Construct a Casimir operator and compute it in a basis.
- Reconstruct the proof of Weyl's complete reducibility theorem.
- Find the equivariant projections of an explicit $\mathfrak{sl}_2$-module.

Throughout, Lie algebras and modules are finite-dimensional over $\mathbb{C}$. Recall from Lecture 4 that $\mathfrak{g}$ is *semisimple* if it has no nonzero solvable ideal.

The route to Weyl's theorem

The main result of this lecture is Weyl's theorem. It says that, for a semisimple Lie algebra, every submodule W of a finite-dimensional module V has a complementary submodule. This subsection is a preview of the proof. It names steps that are proved later, and no proof below uses it.

A complement is the kernel of an intertwining projection. Suppose that an intertwiner $\pi : V \to W$ satisfies $\pi|_W = \mathrm{id}_W$. Then $\ker \pi$ is a submodule, and $V = W \oplus \ker \pi$. Conversely, the projection onto W along a complementary submodule is such an intertwiner. So we look for π.

The maps from V to W form a $\mathfrak{g}$-module $\mathrm{Hom}_{\mathbb{C}}(V, W)$, and its action is defined in § 5.4. Intertwiners are exactly the vectors of this module on which $\mathfrak{g}$ acts by zero. Assume $W \neq 0$, since for $W = 0$ the module V itself is a complement. Inside $\mathrm{Hom}_{\mathbb{C}}(V, W)$ take the maps whose restriction to W is a scalar multiple of id_W, and the smaller space of maps that vanish on W:

$$\mathcal{W} = \{\varphi : \varphi|_W = 0\} \ \subset \ \mathcal{V} = \{\varphi : \varphi|_W \in \mathbb{C}\ \mathrm{id}_W\}.$$

Both are submodules, and $\mathcal{W}$ has codimension one in $\mathcal{V}$. Moreover, $\mathfrak{g}$ maps $\mathcal{V}$ into $\mathcal{W}$. Suppose that a line in $\mathcal{V}$ is a submodule complementary to $\mathcal{W}$. Then $\mathfrak{g}$ maps this line into itself and into $\mathcal{W}$, so $\mathfrak{g}$ acts on it by zero. Hence its nonzero vectors are intertwiners. After rescaling, such a vector restricts to id_W, and it is the map π. The original W need not have codimension one in V. The submodule of codimension one is $\mathcal{W}$, inside $\mathcal{V}$.

So it suffices to split submodules of codimension one. First take an irreducible submodule of codimension one. If $\mathfrak{g}$ acts by zero, any complementary line works. If the action is nonzero,

we construct an operator that commutes with the action, has image in the submodule, and restricts to a nonzero scalar on it. The kernel of this operator is the complement. Next, induction on the dimension treats a submodule of codimension one that is not irreducible.

The commuting operator is the Casimir operator. We introduce the trace form from which it is built in § 5.1, and we construct the operator in § 5.2. We treat codimension one in § 5.3, and we carry out the reduction through $\mathrm{Hom}_{\mathbb{C}}(V, W)$ in § 5.4.

5.1 Trace forms of representations

Definition 5.1 (Trace form). Let $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation. The *trace form* of V is

$$\beta_V(x, y) = \operatorname{tr}\bigl(\rho(x)\rho(y)\bigr), \qquad x, y \in \mathfrak{g}.$$

The trace form is bilinear and symmetric, since $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. For the adjoint module it is the Killing form κ of Lecture 4.

Lemma 5.2 (Invariance · *[P]*). *For all $x, y, z \in \mathfrak{g}$,*

$$\beta_V([x, y], z) = \beta_V(x, [y, z]).$$

Proof. For $A, B, C \in \mathrm{End}(V)$, cyclicity of the trace gives $\operatorname{tr}(BAC) = \operatorname{tr}(ACB)$. Hence

$$\operatorname{tr}([A, B]C) = \operatorname{tr}(ABC) - \operatorname{tr}(BAC) = \operatorname{tr}(ABC) - \operatorname{tr}(ACB) = \operatorname{tr}(A[B, C]).$$

Put $A = \rho(x)$, $B = \rho(y)$, $C = \rho(z)$, and use $\rho([x, y]) = [\rho(x), \rho(y)]$. ∎

Invariance and the antisymmetry of the bracket give the identity used below:

$$\beta_V([z, x], y) = -\beta_V([x, z], y) = -\beta_V(x, [z, y]). \tag{5.1}$$

A trace form can be degenerate. Its radical always contains $\ker \rho$. For semisimple $\mathfrak{g}$, Cartan's criterion shows that the radical is exactly $\ker \rho$.

Theorem 5.3 (Cartan's criterion · *[C]*). *Let V be a finite-dimensional complex vector space and let $L \subseteq \mathfrak{gl}(V)$ be a Lie subalgebra. If*

$$\operatorname{tr}(AB) = 0 \quad \text{for all } A \in [L, L],\ B \in L,$$

then L is solvable.

This is the Theorem in Humphreys, Section 4.3. Lecture 4 stated it. We also use two results proved in Lecture 4. Let $\mathfrak{g}$ be semisimple.

(a) If $\mathfrak{a}$ is an ideal and $\mathfrak{a}^{\perp}$ is its orthogonal space for κ, then $\mathfrak{a}^{\perp}$ is an ideal and $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^{\perp}$.

(b) The algebra $\mathfrak{g}$ is a direct sum of simple ideals $\mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_r$.

Lemma 5.4 (Perfectness · *[P]*). *Let $\mathfrak{g}$ be semisimple. Then $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. For every module V, $\rho(\mathfrak{g}) \subseteq \mathfrak{sl}(V)$. In particular, $\mathfrak{g}$ acts by zero on every one-dimensional module.*

Proof. Each $\mathfrak{g}_i$ in (b) is simple, hence nonabelian. Its derived algebra is a nonzero ideal of $\mathfrak{g}_i$, so $[\mathfrak{g}_i, \mathfrak{g}_i] = \mathfrak{g}_i$. Thus $[\mathfrak{g}, \mathfrak{g}]$ contains every $\mathfrak{g}_i$, and $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$.

Consequently $\rho(\mathfrak{g})$ is spanned by the operators $\rho([x, y]) = [\rho(x), \rho(y)]$, which are traceless. On a one-dimensional space a traceless operator is zero. ∎

Proposition 5.5 (The faithful quotient · *[P]*). *Let $\mathfrak{g}$ be semisimple, let $\rho\colon \mathfrak{g} \to \mathfrak{gl}(V)$ be a representation, and let $\mathfrak{k} = (\ker\rho)^{\perp}$ be the orthogonal space of $\ker\rho$ for κ. Then:*

(i) $\mathfrak{g} = \ker\rho \oplus \mathfrak{k}$ *and* $[\ker\rho, \mathfrak{k}] = 0$*;*

(ii) ρ *restricts to a Lie algebra isomorphism* $\mathfrak{k} \to \rho(\mathfrak{g})$*, so* $\mathfrak{k} \cong \mathfrak{g}/\ker\rho$*;*

(iii) β_V *restricted to* $\mathfrak{k} \times \mathfrak{k}$ *is nondegenerate.*

Proof. Part (i) is result (a) for the ideal $\ker\rho$. The bracket $[\ker\rho, \mathfrak{k}]$ lies in both ideals, and their intersection is zero.

For (ii), $\mathfrak{k} \cap \ker\rho = 0$ gives injectivity. Surjectivity follows from

$$\rho(\mathfrak{g}) = \rho(\ker\rho) + \rho(\mathfrak{k}) = \rho(\mathfrak{k}).$$

For (iii), let

$$S = \{x \in \mathfrak{k} : \beta_V(x, y) = 0 \text{ for all } y \in \mathfrak{k}\}.$$

Let $x \in S$ and $y, z \in \mathfrak{k}$. Since $\mathfrak{k}$ is an ideal, $[z, x]$ and $[z, y]$ lie in $\mathfrak{k}$. Hence (5.1) and $x \in S$ give

$$\beta_V([z, x], y) = -\beta_V(x, [z, y]) = 0.$$

So S is an ideal of $\mathfrak{k}$. By (i), $[\ker\rho, S] = 0$. Hence S is an ideal of $\mathfrak{g}$.

Apply Cartan's criterion to $L = \rho(S)$. For $x, y \in S$,

$$\operatorname{tr}\big(\rho(x)\rho(y)\big) = \beta_V(x, y) = 0.$$

Since $[L, L] \subseteq L$, the hypothesis holds, and $\rho(S)$ is solvable. By (ii), $S \cong \rho(S)$. Thus S is a solvable ideal of $\mathfrak{g}$, so $S = 0$. ∎

The ideal $\mathfrak{k}$ is the faithful quotient $\mathfrak{g}/\ker\rho$, realised inside $\mathfrak{g}$. Since ρ vanishes on $\ker\rho$ and β_V is nondegenerate on $\mathfrak{k}$, the radical of β_V on all of $\mathfrak{g} = \ker\rho \oplus \mathfrak{k}$ is exactly $\ker\rho$.

5.2 The Casimir operator

Keep the notation of Proposition 5.5 and let $m = \dim\mathfrak{k}$. Choose a basis $(x_1, \dots, x_m)$ of $\mathfrak{k}$. Since β_V is nondegenerate on $\mathfrak{k}$, there is a unique *dual basis* $(y_1, \dots, y_m)$ of $\mathfrak{k}$ with

$$\beta_V(x_i, y_j) = \delta_{ij}.$$

Definition 5.6 (Casimir operator). The *Casimir operator* of the module V is

$$c_V = \sum_{i=1}^{m} \rho(x_i)\rho(y_i) \in \operatorname{End}(V).$$

If $\rho = 0$, then $\mathfrak{k} = 0$ and $c_V = 0$. The operator does not depend on the chosen basis. Pairing with y_j, respectively x_j, shows that every $u \in \mathfrak{k}$ satisfies

$$u = \sum_i \beta_V(u, y_i)\, x_i = \sum_i \beta_V(u, x_i)\, y_i.$$

Let (x'_j), (y'_j) be another dual pair. In $\mathfrak{k} \otimes \mathfrak{k}$, expand x'_j by the first rule, then use the second rule for the pair $(x'_j), (y'_j)$:

$$\sum_j x'_j \otimes y'_j = \sum_i x_i \otimes \Big(\sum_j \beta_V(y_i, x'_j)\, y'_j\Big) = \sum_i x_i \otimes y_i.$$

The linear map $\mathfrak{k} \otimes \mathfrak{k} \to \operatorname{End}(V)$, $a \otimes b \mapsto \rho(a)\rho(b)$, turns this into equality of the two operators.

Theorem 5.7 (Casimir operator · *[P]*). *Let $\mathfrak{g}$ be semisimple and let V be a module. Then:*

(i) $c_V \rho(z) = \rho(z) c_V$ *for all* $z \in \mathfrak{g}$*;*

(ii) $\operatorname{tr} c_V = \dim \mathfrak{k}$*;*

(iii) c_V *maps every submodule of V into itself, and if V is irreducible, then*

$$c_V = \frac{\dim \mathfrak{k}}{\dim V}\, \mathrm{id}_V .$$

Proof. If $z \in \ker \rho$, then $\rho(z) = 0$. Since $\mathfrak{g} = \ker \rho \oplus \mathfrak{k}$, it remains to take $z \in \mathfrak{k}$. Since $\mathfrak{k}$ is an ideal, we can write

$$[z, x_i] = \sum_j a_{ij} x_j, \qquad [z, y_i] = \sum_j b_{ij} y_j.$$

Pairing with dual basis vectors and using (5.1) gives

$$a_{ik} = \beta_V([z, x_i], y_k) = -\beta_V(x_i, [z, y_k]) = -b_{ki}.$$

The identity $[A, BC] = [A, B]C + B[A, C]$ in $\operatorname{End}(V)$ now gives

$$\begin{aligned} [\rho(z), c_V] &= \sum_i \big(\rho([z, x_i])\rho(y_i) + \rho(x_i)\rho([z, y_i])\big) \\ &= \sum_{i,j} a_{ij}\, \rho(x_j)\rho(y_i) + \sum_{i,j} b_{ij}\, \rho(x_i)\rho(y_j) \\ &= \sum_{i,j} \big(a_{ji} + b_{ij}\big)\rho(x_i)\rho(y_j) = 0. \end{aligned}$$

The last line renames $i \leftrightarrow j$ in the first sum. The coefficients vanish, since $a_{ji} = -b_{ij}$. This proves (i).

For (ii), linearity of the trace gives

$$\operatorname{tr} c_V = \sum_{i=1}^{m} \beta_V(x_i, y_i) = m = \dim \mathfrak{k}.$$

For (iii), a submodule is preserved by each $\rho(x_i)$ and $\rho(y_i)$, hence by c_V. If V is irreducible, (i) and Schur's lemma make c_V a scalar. Its value follows from (ii) by taking traces. ∎

Remark 5.8 (How the form depends on V). The commutation proof also works with a basis of $\mathfrak{g}$ and its dual for any invariant nondegenerate form on $\mathfrak{g}$. The form β_V, the ideal $\mathfrak{k}$ and c_V depend on V. By Theorem 5.7(ii), the trace of c_V is $\dim \mathfrak{k}$. This is nonzero when $\rho \neq 0$, as needed in Lemma 5.9.

If $W \subseteq V$ is a submodule, then $c_V|_W$ still uses β_V. In general $c_V|_W \neq c_W$ (Exercise 5.3). For a simple algebra all invariant forms are proportional, so changing the form only rescales the Casimir (Exercise 5.2).

5.3 Splitting a codimension-one submodule

We now split a submodule of codimension one. The first lemma uses the Casimir operator, and the second is the induction.

Lemma 5.9 (Irreducible submodule of codimension one · *[P]*). *Let $\mathfrak{g}$ be semisimple, let V be a module, and let $W \subseteq V$ be an irreducible submodule with* $\dim V/W = 1$*. Then $V = W \oplus X$ for some submodule X.*

Proof. If $\rho = 0$, every subspace is a submodule, and any complementary line works. Assume $\rho \neq 0$, so $\mathfrak{k} \neq 0$.

By Lemma 5.4, $\mathfrak{g}$ acts by zero on the one-dimensional module V/W. Thus $\rho(y)V \subseteq W$ for every $y \in \mathfrak{g}$, and therefore

$$c_V(V) \subseteq W.$$

By Theorem 5.7, c_V restricts to an intertwiner $W \to W$. Schur's lemma gives $c_V|_W = \lambda \, \mathrm{id}_W$ for some $\lambda \in \mathbb{C}$.

Choose a basis of W and extend it by one vector of V. In this basis the matrix of c_V is

$$\begin{pmatrix} \lambda I_{\dim W} & * \\ 0 & 0 \end{pmatrix}.$$

Comparing traces with Theorem 5.7(ii) gives

$$\lambda \dim W = \operatorname{tr} c_V = \dim \mathfrak{k} \neq 0.$$

Hence $\lambda \neq 0$.

Put $X = \ker c_V$. It is a submodule, because c_V commutes with $\rho(\mathfrak{g})$. It meets W only in zero, because c_V acts on W by $\lambda \neq 0$. Since the image of c_V lies in W, the rank of c_V is at most $\dim W$, so $\dim X \geq 1$. Since $X \cap W = 0$ and $\dim V/W = 1$, also $\dim X \leq 1$. Hence $\dim X = 1$ and $V = W \oplus X$. ∎

Lemma 5.10 (Codimension one · *[P]*)**.** *Let $\mathfrak{g}$ be semisimple and let $W \subseteq V$ be a submodule with $\dim V/W = 1$. Then $V = W \oplus X$ for some one-dimensional submodule X.*

Proof. Use induction on $\dim W$. The induction hypothesis applies to every module, not only to V. If $W = 0$, take $X = V$. If W is irreducible, apply Lemma 5.9.

Otherwise W has a submodule W' with $0 \neq W' \neq W$. Submodules of V/W' have the form Y/W', where Y is a submodule containing W'. In V/W' the submodule W/W' has codimension one and smaller dimension than W. By induction,

$$V/W' = W/W' \oplus Y/W', \qquad \dim Y/W' = 1.$$

Now W' has codimension one in Y, and $\dim W' < \dim W$. By induction again, $Y = W' \oplus X$ for a one-dimensional submodule X.

If $v \in Y \cap W$, then $v + W'$ lies in $Y/W' \cap W/W' = 0$, so $v \in W'$. Hence

$$X \cap W = X \cap Y \cap W \subseteq X \cap W' = 0.$$

Since $\dim W + \dim X = \dim V$, we get $V = W \oplus X$. ∎

When $W \neq 0$, the induction ends in Lemma 5.9 for some subquotient of V. Each step with nonzero action uses the Casimir operator of that subquotient, formed with its own trace form.

5.4 Weyl's theorem

We now carry out the reduction to codimension one. First we need the module $\operatorname{Hom}_{\mathbb{C}}(V, W)$.

For modules V and W, the space $\operatorname{Hom}_{\mathbb{C}}(V, W)$ is a $\mathfrak{g}$-module with

$$x \cdot \varphi = \rho_W(x)\varphi - \varphi\rho_V(x).$$

Expand $x \cdot (y \cdot \varphi)$ and $y \cdot (x \cdot \varphi)$. In the difference, the four mixed terms such as $\rho_W(x)\varphi\rho_V(y)$ cancel in pairs. Since ρ_V and ρ_W preserve brackets,

$$\begin{aligned} x \cdot (y \cdot \varphi) - y \cdot (x \cdot \varphi) &= [\rho_W(x), \rho_W(y)]\varphi - \varphi[\rho_V(x), \rho_V(y)] \\ &= [x, y] \cdot \varphi. \end{aligned}$$

By definition, $x \cdot \varphi = 0$ for all x exactly when $\varphi \in \operatorname{Hom}_{\mathfrak{g}}(V, W)$.

Theorem 5.11 (Weyl's theorem · *[P]*)**.** *Let $\mathfrak{g}$ be semisimple and let V be a finite-dimensional module. Every submodule $W \subseteq V$ has a complementary submodule. Consequently V is completely reducible.*

Proof. If $W = 0$, take V itself. Assume $W \neq 0$. Inside $\operatorname{Hom}_{\mathbb{C}}(V, W)$ consider

$$\mathcal{V} = \{\varphi : \varphi|_W \in \mathbb{C}\ \mathrm{id}_W\}, \qquad \mathcal{W} = \{\varphi : \varphi|_W = 0\}.$$

Step 1: $\mathfrak{g} \cdot \mathcal{V} \subseteq \mathcal{W}$. Let $\varphi|_W = s\ \mathrm{id}_W$ and $w \in W$. Since $\rho_V(x)w \in W$,

$$(x \cdot \varphi)(w) = \rho_W(x)(sw) - \varphi\big(\rho_V(x)w\big) = s\rho_W(x)w - s\rho_V(x)w = 0.$$

As $\mathcal{W} \subseteq \mathcal{V}$, both $\mathcal{V}$ and $\mathcal{W}$ are submodules of $\operatorname{Hom}_{\mathbb{C}}(V, W)$.

Step 2: codimension one. The map $\mathcal{V} \to \mathbb{C}$, $\varphi \mapsto s$, is linear with kernel $\mathcal{W}$. It is onto, because a linear projection of V onto W along any complementary subspace has $s = 1$.

Step 3: an invariant projection. By Lemma 5.10, $\mathcal{V} = \mathcal{W} \oplus \mathbb{C}\pi$ with $\mathbb{C}\pi$ a submodule. For each x, Step 1 gives $x \cdot \pi \in \mathcal{W} \cap \mathbb{C}\pi = 0$. Since $\pi \notin \mathcal{W}$, we may rescale so that $\pi|_W = \mathrm{id}_W$. Thus $\pi \in \operatorname{Hom}_{\mathfrak{g}}(V, W)$ is a projection onto W.

Step 4: the complement. Put $X = \ker \pi$. It is a submodule, since π is an intertwiner. Each $v \in V$ satisfies

$$v = \pi(v) + \big(v - \pi(v)\big), \qquad \pi\big(v - \pi(v)\big) = \pi(v) - \pi(v) = 0,$$

because $\pi(v) \in W$ and $\pi|_W = \mathrm{id}_W$. Also $X \cap W = 0$. Hence $V = W \oplus X$.

Complete reducibility follows by induction on $\dim V$. The zero module is the empty direct sum, and an irreducible module is its own decomposition. Otherwise choose a submodule W with $0 \neq W \neq V$. Then $V = W \oplus X$, and W and X are modules of smaller dimension. ∎

The proof uses linear algebra, Cartan's criterion, Schur's lemma and the two results on ideals quoted from Lecture 4. An algebraic proof of this kind goes back to Casimir and van der Waerden.

Semisimplicity entered twice. It gave the nondegenerate trace form of Proposition 5.5, and the trivial action on one-dimensional modules in Lemma 5.4.

5.5 Worked example: splitting $\operatorname{End}(\mathbb{C}^2)$

Let $\mathfrak{sl}_2$ act on $\mathfrak{gl}_2 = \operatorname{End}(\mathbb{C}^2)$ by

$$\rho(x)A = [x, A] = xA - Ax.$$

This is the module $\operatorname{Hom}_{\mathbb{C}}(\mathbb{C}^2, \mathbb{C}^2)$ of § 5.4, with both arguments the defining module. Two submodules are visible:

$$\mathbb{C}I_2, \quad \text{since } [x, I_2] = 0, \qquad \mathfrak{sl}_2, \quad \text{since } \operatorname{tr}[x, A] = 0.$$

On $\mathfrak{sl}_2$ the action is the adjoint action. Every matrix $A \in \mathfrak{gl}_2$ splits as

$$A = \frac{1}{2}(\operatorname{tr} A)\, I_2 + \Big(A - \frac{1}{2}(\operatorname{tr} A)\, I_2\Big).$$

The projection in matrix form

Use the ordered basis $(E_{11}, E_{12}, E_{21}, E_{22})$ of $\mathfrak{gl}_2$. The brackets

$$[e, E_{11}] = -E_{12}, \quad [e, E_{12}] = 0, \quad [e, E_{21}] = E_{11} - E_{22}, \quad [e, E_{22}] = E_{12}$$

give

$$D(e) = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix}.$$

The coordinate vector of I_2 is $u = (1, 0, 0, 1)^{\mathsf{T}}$. The trace of A is the row u^{T} applied to the coordinates of A. The projection $P(A) = \frac{1}{2}(\operatorname{tr} A) I_2$ onto $\mathbb{C}I_2$ along $\mathfrak{sl}_2$ therefore has matrix

$$D(P) = \frac{1}{2} u u^{\mathsf{T}} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}.$$

For e the check is direct:

$$D(e)u = 0, \qquad u^{\mathsf{T}} D(e) = (0, 0, 0, 0).$$

Hence $D(P)D(e) = 0 = D(e)D(P)$. The same holds for every $x \in \mathfrak{sl}_2$. Indeed, $D(x)u = 0$ says $[x, I_2] = 0$, and $u^{\mathsf{T}} D(x) = 0$ says $\operatorname{tr}[x, A] = 0$ for every A. So P, viewed as a map $\mathfrak{gl}_2 \to \mathbb{C}I_2$, gives the invariant projection in Step 3 of Theorem 5.11. The complement is $\ker P = \mathfrak{sl}_2$.

The Casimir operator

Write c for $c_{\mathfrak{gl}_2}$. In the ordered basis (I_2, h, e, f) of $\mathfrak{gl}_2$, each $\rho(x)$ is block diagonal, with block 0 on $\mathbb{C}I_2$ and block $\operatorname{ad} x$ on $\mathfrak{sl}_2$. Hence the trace form of $\mathfrak{gl}_2$ is the Killing form of $\mathfrak{sl}_2$. By Lecture 4,

$$\kappa(h, h) = 8, \qquad \kappa(e, f) = \kappa(f, e) = 4,$$

and κ vanishes on the other pairs of basis vectors. The module is faithful, so $\mathfrak{k} = \mathfrak{sl}_2$. The dual basis of (h, e, f) is $(h/8,\ f/4,\ e/4)$, and

$$c = \frac{1}{8}\rho(h)^2 + \frac{1}{4}\rho(e)\rho(f) + \frac{1}{4}\rho(f)\rho(e).$$

On h, using $[f, h] = 2f$ and $[e, h] = -2e$,

$$\begin{aligned} c(h) &= \frac{1}{8}[h, [h, h]] + \frac{1}{4}[e, [f, h]] + \frac{1}{4}[f, [e, h]] \\ &= \frac{1}{4}[e, 2f] + \frac{1}{4}[f, -2e] = \frac{1}{2}h + \frac{1}{2}h = h. \end{aligned}$$

On e and f, using $[h, [h, e]] = 4e$ and $[h, [h, f]] = 4f$,

$$\begin{aligned} c(e) &= \frac{1}{8} \cdot 4e + \frac{1}{4}[e, -h] = \frac{1}{2}e + \frac{1}{2}e = e, \\ c(f) &= \frac{1}{8} \cdot 4f + \frac{1}{4}[f, h] = \frac{1}{2}f + \frac{1}{2}f = f. \end{aligned}$$

Also $c(I_2) = 0$. In the basis (I_2, h, e, f) the matrix of c is

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Its trace is $3 = \dim \mathfrak{k}$, as Theorem 5.7(ii) predicts. The submodule $\mathfrak{sl}_2$ is irreducible, since its submodules are ideals of the simple algebra $\mathfrak{sl}_2$. The two equivariant projections are

$$c \colon \mathfrak{gl}_2 \to \mathfrak{sl}_2, \qquad \mathrm{id} - c = P \colon \mathfrak{gl}_2 \to \mathbb{C}I_2.$$

This is Lemma 5.9 for $W = \mathfrak{sl}_2$, with $\lambda = 1$ and $X = \ker c = \mathbb{C}I_2$.

Contrast with Lecture 3

The faithful module of the solvable algebra $\mathfrak{b} = \mathbb{C}h \oplus \mathbb{C}e$, $[h, e] = e$, in Lecture 3 has a degenerate trace form. The algebra is not perfect either. Exercise 5.5 computes these failures and the corresponding Hom-space obstruction.

Self-checks

(1) For the defining $\mathfrak{sl}_2$-module, $\beta_V(x, y) = \mathrm{tr}(xy)$. Find the dual basis of (h, e, f) and the Casimir operator.

(2) Why does a semisimple algebra act by zero on every one-dimensional module, while $\mathfrak{b}$ above does not?

(3) Let $\mathbb{C}t$ act on $\mathbb{C}^2$ with $D(t) = E_{12}$ in the standard basis, as in Lecture 1. Does this contradict Weyl's theorem? What does its trace form detect?

Answers. (1) The values $\mathrm{tr}(h^2) = 2$ and $\mathrm{tr}(ef) = \mathrm{tr}(fe) = 1$, with all other pairings zero, give the dual basis $(h/2, f, e)$. Then

$$c_V = \frac{1}{2}h^2 + ef + fe = \frac{1}{2}I_2 + E_{11} + E_{22} = \frac{3}{2}I_2,$$

in agreement with $\dim \mathfrak{k}/\dim V = 3/2$. (2) A semisimple algebra satisfies $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$, and commutators act by zero on a line. For $\mathfrak{b}$, $[\mathfrak{b}, \mathfrak{b}] = \mathbb{C}e$, and h may act by any scalar. (3) No. The algebra $\mathbb{C}t$ is abelian, hence solvable and not semisimple. The module is faithful, but $\mathrm{tr}\, D(t)^2 = 0$, so the trace form is zero. By Proposition 5.5, this also rules out semisimplicity.

Summary

A module V carries an invariant trace form. By Cartan's criterion, for semisimple $\mathfrak{g}$ this form is nondegenerate on the faithful quotient $\mathfrak{k} \cong \mathfrak{g}/\ker \rho$. The dual-basis Casimir c_V commutes with the action and has trace $\dim \mathfrak{k}$.

For a nonzero action, the kernel of the Casimir complements an irreducible submodule of codimension one. For a zero action, any complement is invariant. Induction treats every submodule of codimension one. The module $\mathrm{Hom}_{\mathbb{C}}(V, W)$ reduces an arbitrary submodule to that case. Hence finite-dimensional modules of semisimple algebras are completely reducible. Lecture 6 uses this to decompose $\mathfrak{sl}_2$-modules and their tensor products.

Reading

Humphreys, *Introduction to Lie Algebras and Representation Theory*, Sections 6.1–6.3, and Section 4.3 for Cartan's criterion. Erdmann–Wildon, *Introduction to Lie Algebras*, Appendix B. Exercise Sheet 5 computes further Casimir operators and commutants.